

Streamflow Forecasting Using ANN and LSTM

Machine Learning Mini-Project — UE24CS352A

1. Problem Statement

Streamflow forecasting supports water-resource management, flood prediction, and hydrological planning. This project investigates forecasting future streamflow from historical rainfall and streamflow using an Artificial Neural Network (ANN) and a Long Short-Term Memory (LSTM) network. The approach follows the lag-based methodology of the reference study, where current runoff is modeled from previous rainfall and runoff observations.

2. Dataset and Preprocessing

The dataset contains daily rainfall and streamflow observations for 2003–2012. Data from 2003–2007 were used for training and 2008–2012 for validation. Rainfall was converted from mm/day to inches/day, missing rainfall values were forward-filled, and three previous rainfall and three previous streamflow observations were used as lag features. Values were normalized using the ranges adopted from the reference methodology.

3. Methodology

ANN: A feed-forward network with a 32-unit ReLU hidden layer and linear output was trained with Adam, learning rate 0.001, 30 epochs, and batch size 32.

LSTM: Lagged rainfall and streamflow values were arranged as a three-step sequence with two features per step. An LSTM layer with 32 units was followed by a linear output layer.

Evaluation: RMSE, MAE, and R^2 were calculated on validation predictions. Peak and non-peak behavior was evaluated using 1500 cfs as the threshold. ANN hyperparameters were also varied across hidden sizes 16/32 and learning rates 0.001/0.005.

4. Implementation

The notebook implements data loading, preprocessing, lag generation, normalization, ANN and LSTM training, prediction, hydrograph visualization, evaluation, peak/non-peak analysis, and hyperparameter comparison. Result tables are exported as CSV files and hydrographs as PNG files.

5. Overall Results

Model	RMSE (cfs)	MAE (cfs)	R^2
ANN	539.84	307.43	0.9076
LSTM	636.51	260.39	0.8715

The ANN achieved lower RMSE and higher R^2 than the LSTM in the overall validation evaluation, while the LSTM had lower MAE.

6. Peak and Non-Peak Analysis

Model	Region	RMSE (cfs)	MAE (cfs)
ANN	Peak	1343.03	913.64
ANN	Non-peak	280.41	222.12
LSTM	Peak	1711.36	1117.45
LSTM	Non-peak	223.69	139.79

Both models show substantially larger errors during peak events than during non-peak periods, making extreme flows the main forecasting challenge.

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7. Hyperparameter Study

Hidden Units	Learning Rate	RMSE (cfs)
32	0.001	513.83
16	0.001	537.06
32	0.005	565.00
16	0.005	573.44

The best observed hyperparameter run used 32 hidden units and learning rate 0.001, with validation RMSE 513.83 cfs. Neural-network training is stochastic, so repeated runs can vary slightly.

8. Comparison and Discussion

The ANN performed better overall according to RMSE and R^2 . For the available lag-based representation, the additional sequential modeling capability of the LSTM did not improve overall accuracy. The LSTM nevertheless produced lower errors on non-peak flows. Both models struggled more with high-flow events, where accurate prediction is more difficult.

9. Conclusions

The project demonstrates a complete machine-learning workflow for daily streamflow forecasting from historical rainfall and streamflow. Both ANN and LSTM were implemented and compared. The ANN obtained RMSE 539.84 cfs and R^2 0.9076, compared with LSTM RMSE 636.51 cfs and R^2 0.8715. Peak events were substantially harder to predict. The best tested ANN hyperparameter configuration was 32 hidden units with learning rate 0.001.

10. Limitations and Future Scope

The current model uses a limited set of lagged rainfall and streamflow features. Future work can include additional environmental or geographical variables, longer lag windows, peak-aware training, and more systematic hyperparameter tuning.

11. Repository and Reproducibility

The GitHub repository contains the main notebook, evaluation CSV files, hydrograph visualizations, and README documentation. The notebook can be executed in Google Colab or Jupyter after providing the required dataset.